ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P2H Worked Solutions

Jan 2022 | P2H

33 questions ▪ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

STUDENT EDITION

Worked answers included

DOWNLOAD

eliteigcse.com

PAST PAPER QUESTION Q#: 1 Marks: 4

Topic: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

1 (a) Expand and simplify $(y + 4)(2 - y)$

....

(2)

(b) Factorise fully $15b^5c - 35b^3c^9$

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Expanding brackets. The tag is correct.

Solution

(a)

$$(y+4)(2-y) = 2y - y^2 + 8 - 4y = -y^2 - 2y + 8$$

(b)

$$15b^5c - 35b^3c^9 = 5b^3c(3b^2 - 7c^8)$$

FINAL ANSWER

(a) $-y^2 - 2y + 8$, (b) $5b^3c(3b^2 - 7c^8)$

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Fractions. The tag is correct.

Method

Convert to improper fractions:

$$6\frac{3}{4} = \frac{27}{4}$$

$$2\frac{4}{7} = \frac{18}{7}$$

Divide by multiplying by the reciprocal:

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

$$= \frac{21}{8} = 2\frac{5}{8}$$

FINAL ANSWER

$2\frac{5}{8}$

PAST PAPER QUESTION Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

3

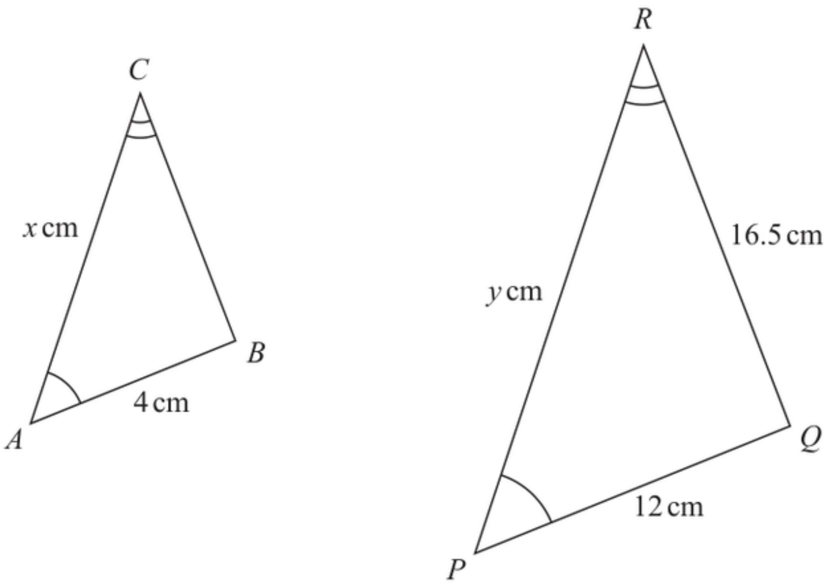


Diagram **NOT** accurately drawn

Triangle ABC is similar to triangle PQR

$AB = 4$ cm $\quad PQ = 12$ cm $\quad RQ = 16.5$ cm $\quad AC = x$ cm $\quad PR = y$ cm

(a) Calculate the length of BC

.... cm

(2)

(b) Write down an expression for y in terms of x

$y =$ ....

(1)

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Congruence, Similarity & Geometrical Proof. The tag is correct.

Solution

The scale factor from triangle ABC to triangle PQR is

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

So

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Also PR corresponds to AC, so

$$y = 3x$$

FINAL ANSWER

$BC = 5.5$ cm, and $y = 3x$.

PAST PAPER QUESTION Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

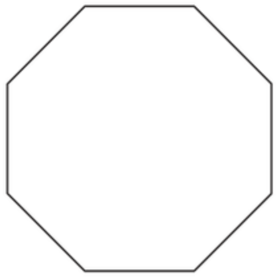


Diagram **NOT** accurately drawn

(a) Write down the lower bound of the length of each side of the octagon.

.... mm

(1)

(b) Write down the upper bound of the length of each side of the octagon.

.... mm

(1)

(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The side length is 18 mm correct to the nearest 0.5 mm.
Half of 0.5 mm is 0.25 mm.

$$18 - 0.25 = 17.75$$

$$18 + 0.25 = 18.25$$

FINAL ANSWER

(a) 17.75 mm, (b) 18.25 mm

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

5 The scale diagram shows the position on a map of a house, A

Scale: 1 cm represents 200 metres

House C is on a bearing of 110° from A
The distance from A to C is 700 m

(a) Mark the position of C on the diagram with a cross (×)
Label your cross C

(3)

(b) Write the scale of the map in the form $1 : n$

1 : ....

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Solution

The distance from A to C is 700 m.
The scale is

$$1 \text{ cm represents } 200 \text{ m}$$

So the map distance is

$$700 \div 200 = 3.5 \text{ cm}$$

Part (a): from A, draw a bearing of 110° and mark C at 3.5 cm from A.
For part (b),

$$200 \text{ m} = 20000 \text{ cm}$$

So the scale is

$$1 : 20000$$

FINAL ANSWER

Draw C 3.5 cm from A on a bearing of 110°, and the scale is $1 : 20000$.

PAST PAPER QUESTION Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio
number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Probability Toolkit. The ratio is used, but the main structure is a probability table.

Method

The probabilities for green and red are

$$0.20 + 0.35 = 0.55$$

So

$$P(\text{pink or white}) = 1 - 0.55 = 0.45$$

Pink : white $= 2 : 1$, so white is $\frac{1}{3}$ of 0.45.

$$P(\text{white}) = \frac{1}{3} \times 0.45 = 0.15$$

The red probability is 0.35, and there are 28 red sweets.

$$\text{total sweets} = \frac{28}{0.35} = 80$$

Number of white sweets:

$$0.15 \times 80 = 12$$

FINAL ANSWER

12

PAST PAPER QUESTION

Topic: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 7 Marks: 3

7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$28 = 2^2 \times 7$$

$$42 = 2 \times 3 \times 7$$

$$63 = 3^2 \times 7$$

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2^2 \times 3^2 \times 7 = 252$$

FINAL ANSWER

252

PAST PAPER QUESTION Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

8 The table gives information about the average house price in England in 2018 and in 2019

Year	**2017**	**2018**	**2019**
Average house price (£)		228 314	231 776

(a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

.... %
(2)

The average house price in 2019 was 7.7% greater than the average house price in 2017

(b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£ ....
(3)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Percentages. The tag is correct.

Method

Increase from 2018 to 2019:

$$231776 - 228314 = 3462$$

Percentage increase:

$$\frac{3462}{228314} \times 100 = 1.516\ldots$$

Correct to one decimal place:

$$1.5\%$$

For part (b), the 2019 price was 7.7% greater than the 2017 price.

$$231776 = 1.077 \times 2017 \text{ price}$$

$$2017 \text{ price} = \frac{231776}{1.077} = 215205.19\ldots$$

Correct to 3 significant figures:

$$215000$$

FINAL ANSWER

(a) 1.5%, (b) £215000

PAST PAPER QUESTION

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 9 Marks: 4

9 The frequency table gives information about the number of points scored by a player.

Number of points	Frequency
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$ ....

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

The total number of points is

$$0(13) + 1(17) + 2(8) + 3x + 4(11) = 77 + 3x$$

The total frequency is

$$13 + 17 + 8 + x + 11 = 49 + x$$

The mean is 2, so

$$\frac{77 + 3x}{49 + x} = 2$$

$$77 + 3x = 98 + 2x$$

$$x = 21$$

FINAL ANSWER

$x = 21$.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jan 2022 P2H - Jan 2022 | P2H

10 Solve the simultaneous equations

$$3x + 5y = 3.1$$
$$6x + 3y = 3.75$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous equations. The tag is correct.

Solution

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Double the first equation:

$$6x + 10y = 6.2$$

Subtract the second equation:

$$7y = 2.45$$

$$y = 0.35$$

Substitute into $3x + 5y = 3.1$:

$$3x + 5(0.35) = 3.1$$

$$3x + 1.75 = 3.1$$

$$3x = 1.35$$

$$x = 0.45$$

FINAL ANSWER

$x = 0.45$, $y = 0.35$.

PAST PAPER QUESTION

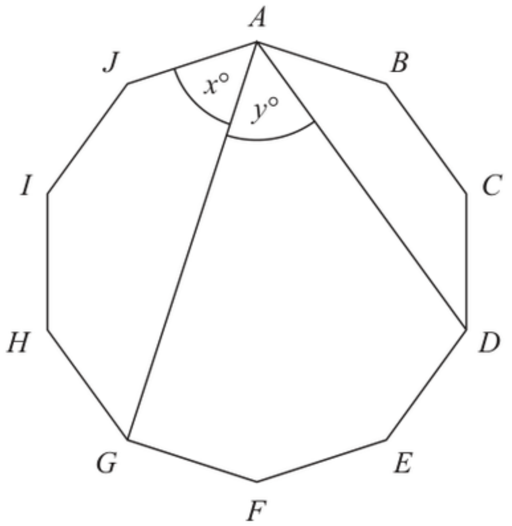
Q#: 11 **Marks: 4**

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

11 The diagram shows a regular 10-sided polygon, *ABCDEFGHIJ*

Diagram **NOT** accurately drawn



Show that $x = y$

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

A regular 10-sided polygon can be divided into 10 equal arcs around its circumcircle.
The angle x subtends the arc from J to G, which covers 3 equal sides of the decagon.
The angle y subtends the arc from G to D, which also covers 3 equal sides of the decagon.
Equal arcs subtend equal angles at the circumference, so

$$x = y$$

FINAL ANSWER

$x = y$.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Powers, roots and standard form. The tag is correct.

Method

$$a = 6 \times 10^{40}$$

$$a^3 = (6 \times 10^{40})^3$$

$$a^3 = 6^3 \times 10^{120}$$

$$a^3 = 216 \times 10^{120}$$

Write this in standard form:

$$216 \times 10^{120} = 2.16 \times 10^{122}$$

FINAL ANSWER

2.16×10^{122}

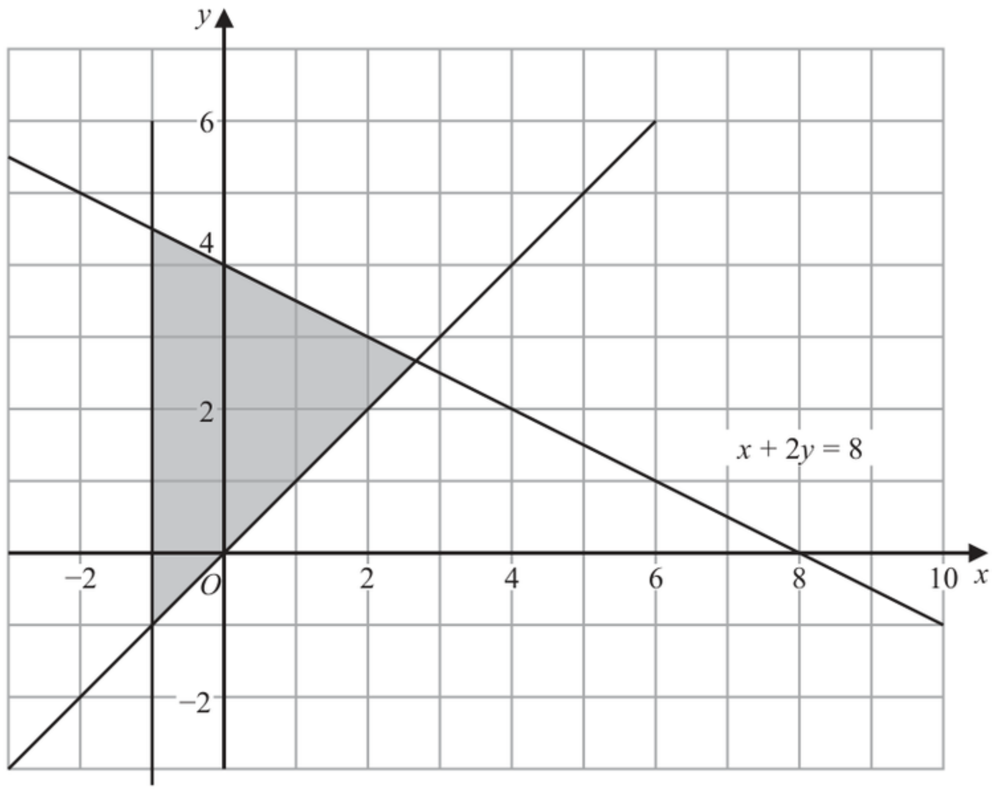
PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct. This is identifying inequalities from a shaded region.

Solution

The region is to the right of the y-axis:

$$x \geq 0$$

It is above the line $y = x$:

$$y \geq x$$

It is below the line

$$x + 2y = 8$$

so

$$x + 2y \leq 8$$

FINAL ANSWER

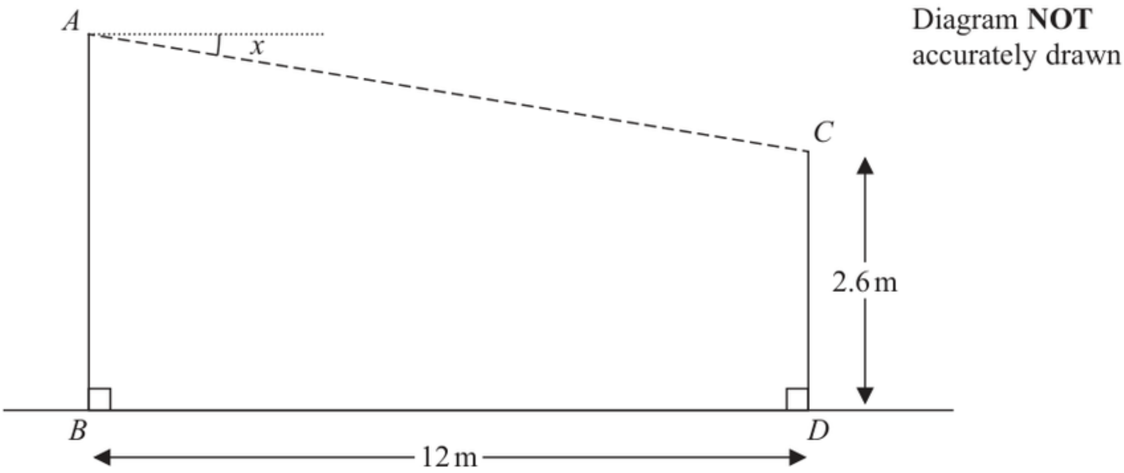
$x \geq 0$, $y \geq x$, $x + 2y \leq 8$.

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

14 A zip wire is shown as the dashed line *AC* in the diagram.



The zip wire is supported by two vertical posts *AB* and *CD* standing on horizontal ground.

$CD = 2.6\,\text{m} \qquad BD = 12\,\text{m}$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post *AB*
Give your answer correct to 3 significant figures.

.... m

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION Q#: 14 Marks: 3

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

The horizontal distance is 12 m and the zip wire is 13 m.
Vertical difference:

$$\sqrt{13^2 - 12^2} = 5$$

Since the shorter post is 2.6 m,

$$AB = 2.6 + 5 = 7.6$$

FINAL ANSWER
7.60 m.

PAST PAPER QUESTION Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8	10	12	13	5	23	21	7	5	16	14

Find the interquartile range of his results.

.... km

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

Put the distances in order:

$$5,\ 5,\ 7,\ 8,\ 10,\ 12,\ 13,\ 14,\ 16,\ 21,\ 23$$

There are 11 values, so the median is the 6th value.
The lower half is

$$5,\ 5,\ 7,\ 8,\ 10$$

so

$$Q_1 = 7$$

The upper half is

$$13,\ 14,\ 16,\ 21,\ 23$$

so

$$Q_3 = 16$$

$$\text{IQR} = 16 - 7 = 9$$

FINAL ANSWER

9 km.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

16 D, E, F and G are points on a circle, centre O

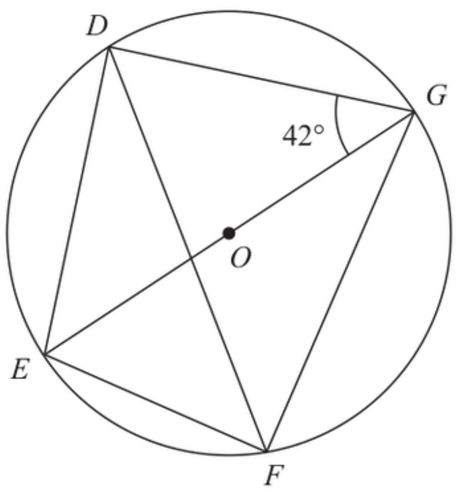


Diagram **NOT** accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42°$

Calculate the size of angle DFG
Give a reason for each stage of your working.

Angle DFG = ... °

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circle Theorems. The tag is correct.

Solution

Since EOG is a diameter,

$$\angle EDG = 90^\circ$$

In triangle EDG,

$$\angle DEG = 180^\circ - 90^\circ - 42^\circ = 48^\circ$$

Angles DEG and DFG stand on the same chord DG, so they are equal.

$$\angle DFG = 48^\circ$$

FINAL ANSWER

48°.

PAST PAPER QUESTION

Q#: 17 **Marks: 3**

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

17 Show that $\dfrac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a+\sqrt{b}$ where a and b are integers.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks for an exact-form proof.

Solution

$$\frac{\sqrt{12}}{\sqrt{3}+2} = \frac{2\sqrt{3}}{\sqrt{3}+2}$$

Multiply by the conjugate:

$$\frac{2\sqrt{3}}{\sqrt{3}+2} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2\sqrt{3}(2-\sqrt{3})}{4-3}$$

$$= 4\sqrt{3} - 6$$

Since $4\sqrt{3} = \sqrt{48}$,

$$4\sqrt{3} - 6 = -6 + \sqrt{48}$$

FINAL ANSWER

$-6 + \sqrt{48}$, so $a = -6$ and $b = 48$.

PAST PAPER QUESTION Q#: 18 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is correct.

Solution

Let the two consecutive odd numbers be

$$2n + 1 \quad \text{and} \quad 2n + 3$$

The sum of their squares is

$$(2n + 1)^2 + (2n + 3)^2$$

$$= 4n^2 + 4n + 1 + 4n^2 + 12n + 9$$

$$= 8n^2 + 16n + 10$$

$$= 8(n^2 + 2n + 1) + 2 = 8(n + 1)^2 + 2$$

This is a multiple of 8 plus 2, so the remainder is always 2.

FINAL ANSWER

The remainder is always 2.

PAST PAPER QUESTION

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 19 Marks: 3

19

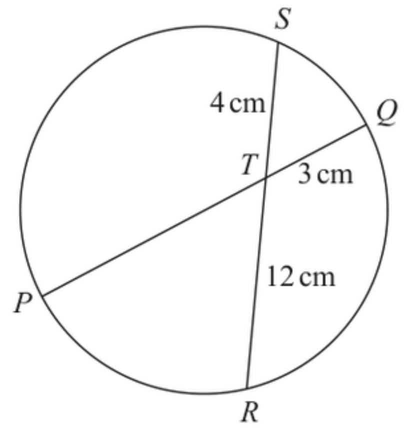


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3$ cm $\qquad ST = 4$ cm $\qquad TR = 12$ cm

Calculate the radius of the circle.

.... cm

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

Topic: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circle Theorems. The tag is correct.

Solution

Use the intersecting chords theorem:

$$PT \times TQ = ST \times TR$$

$$PT \times 3 = 4 \times 12$$

$$PT = 16$$

Since PTQ is a diameter,

$$PQ = PT + TQ = 16 + 3 = 19$$

So the radius is

$$19 \div 2 = 9.5$$

FINAL ANSWER

9.5 cm.

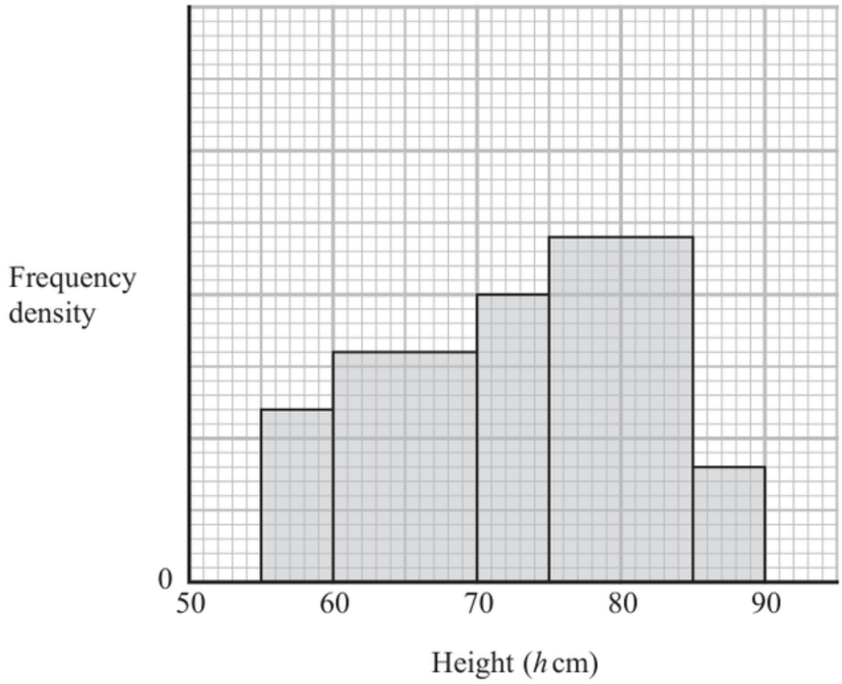
PAST PAPER QUESTION

Q#: 20 **Marks: 4**

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

FINAL ANSWER

$\frac{1}{6}$

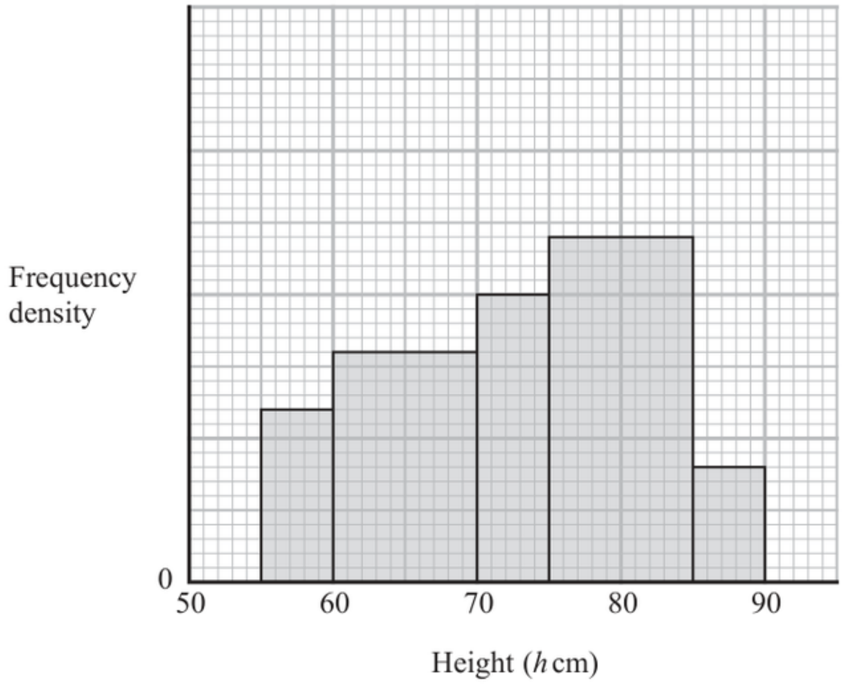
PAST PAPER QUESTION

Q#: 20 Marks: 4

Topic: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

FINAL ANSWER

$\frac{1}{6}$

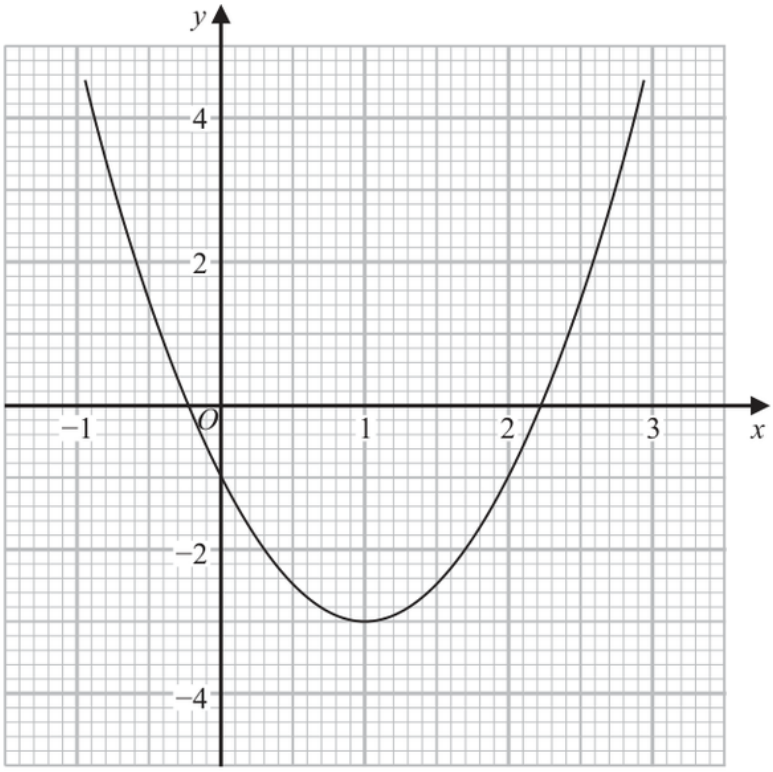
PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Graphs of Functions.

Method

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

The given graph is

$$y = 2x^2 - 4x - 1$$

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

So draw the straight line

$$y = 1 - 2x$$

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

FINAL ANSWER

(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

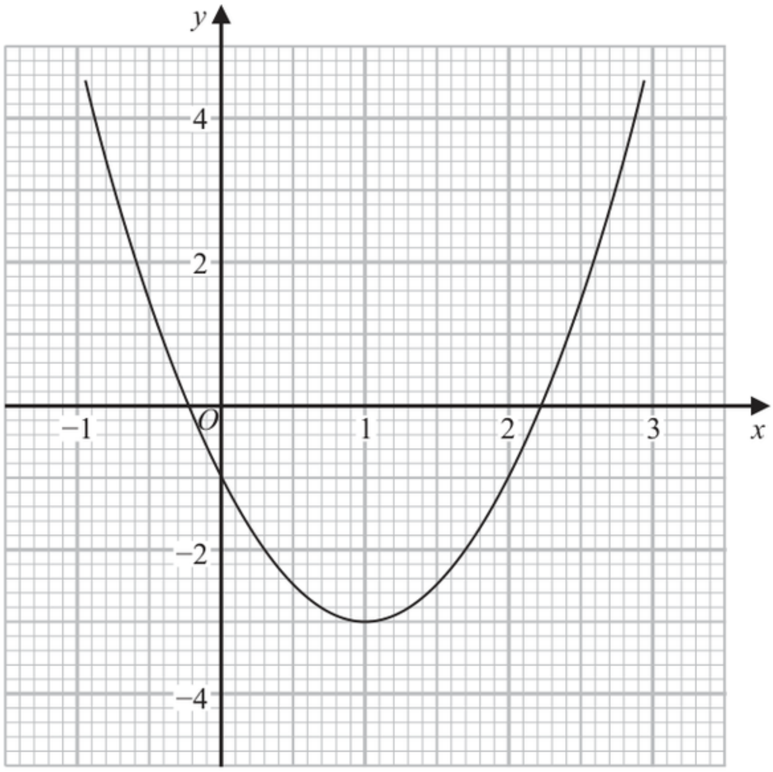
PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Graphs of Functions.

Method

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

The given graph is

$$y = 2x^2 - 4x - 1$$

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

So draw the straight line

$$y = 1 - 2x$$

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

FINAL ANSWER

(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

22 Here is a rectangle.

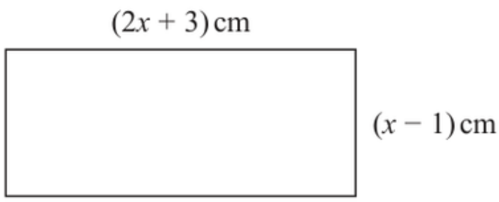


Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Solving Inequalities.

Method

The rectangle has side lengths

$$2x + 3$$

and

$$x - 1$$

For the side lengths to be possible,

$$x > 1$$

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

So

$$-\frac{13}{2} < x < 6$$

Combining with $x > 1$:

$$1 < x < 6$$

FINAL ANSWER

$1 < x < 6$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

22 Here is a rectangle.

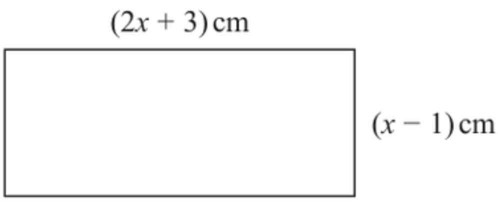


Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Solving Inequalities.

Method

The rectangle has side lengths

$$2x + 3$$

and

$$x - 1$$

For the side lengths to be possible,

$$x > 1$$

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

So

$$-\frac{13}{2} < x < 6$$

Combining with $x > 1$:

$$1 < x < 6$$

FINAL ANSWER

$1 < x < 6$

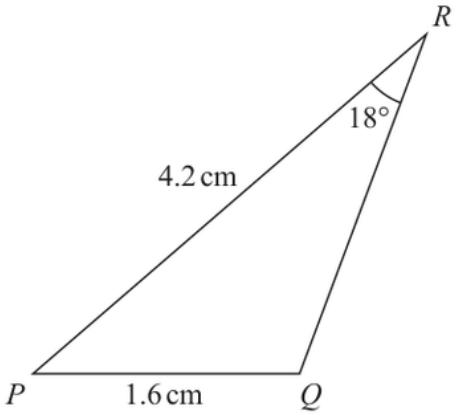
PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle *PQR*

Diagram **NOT** accurately drawn



$PQ = 1.6$ cm $\quad PR = 4.2$ cm $\quad$ Angle $PRQ = 18°$

Given that angle *PQR* is obtuse,

work out the area of triangle *PQR*
Give your answer correct to 3 significant figures.

.... cm²

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

Correct to 3 significant figures:

$$1.98$$

FINAL ANSWER

$\boxed{1.98 \text{ cm}^2}$

PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle PQR

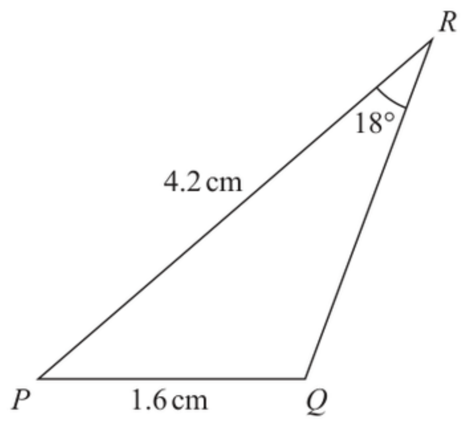


Diagram **NOT** accurately drawn

$PQ = 1.6\,\text{cm}$ $PR = 4.2\,\text{cm}$ Angle $PRQ = 18°$

Given that angle PQR is obtuse,

work out the area of triangle PQR
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

Correct to 3 significant figures:

$$1.98$$

FINAL ANSWER

$\boxed{1.98 \text{ cm}^2}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Differentiation.

Method

$$x = 4t^3 - 27t + 8$$

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

Since $t \geq 0$, $t = \frac{3}{2}$.
Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

At $t = \frac{3}{2}$,

$$a = 24\left(\frac{3}{2}\right) = 36$$

FINAL ANSWER
$a = 36$

PAST PAPER QUESTION Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Differentiation.

Method

$$x = 4t^3 - 27t + 8$$

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

Since $t \geq 0$, $t = \frac{3}{2}$.
Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

At $t = \frac{3}{2}$,

$$a = 24\left(\frac{3}{2}\right) = 36$$

FINAL ANSWER
$a = 36$

PAST PAPER QUESTION **Q#: 25** **Marks: 5**

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain} \quad \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$g(x) = 5 + 6x - x^2$$

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

Let

$$y = 14 - (x - 3)^2$$

$$(x - 3)^2 = 14 - y$$

The domain of g is $x \geq 3$, so take the positive square root:

$$x - 3 = \sqrt{14 - y}$$

$$x = 3 + \sqrt{14 - y}$$

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

The range of g is $x \leq 14$, so this is the domain of g^{-1}.

FINAL ANSWER

(a) $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$, (b) domain $x \leq 14$

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain} \quad \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$g(x) = 5 + 6x - x^2$$

Complete the square:

$$g(x) = 14 - (x-3)^2$$

Let

$$y = 14 - (x-3)^2$$

$$(x-3)^2 = 14 - y$$

The domain of g is $x \geq 3$, so take the positive square root:

$$x - 3 = \sqrt{14-y}$$

$$x = 3 + \sqrt{14-y}$$

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14-x}$$

The range of g is $x \leq 14$, so this is the domain of g^{-1}.

FINAL ANSWER

(a) $g^{-1} : x \mapsto 3 + \sqrt{14-x}$, (b) domain $x \leq 14$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$
$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

d = ...

m = ...

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$S_m = 39$$

$$S_{2m} = 320$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Then

$$2S_m = m(2a + (m-1)d) = 78$$

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2 d$$

$$242 = 2 \times 11^2$$

Since d is prime and m^2 is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

FINAL ANSWER

$d = 2,\ m = 11$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$
$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$S_m = 39$$

$$S_{2m} = 320$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Then

$$2S_m = m(2a + (m-1)d) = 78$$

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2 d$$

$$242 = 2 \times 11^2$$

Since d is prime and m^2 is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

FINAL ANSWER

$d = 2, \ m = 11$